# Koustav Mallick

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## Technical Skills

**Cloud & Infra: GCP (BigQuery, Dataproc, Dataflow, Cloud Run), AWS (DynamoDB, Lambda, S3, EC2), Terraform, Docker**
**Big Data & AI: Apache Spark, PySpark, ETL, Hadoop, Hive, Kafka Streaming, MCP, TensorFlow, PyTorch, Transformers**
**Backend & APIs: Spring Boot, FastAPI, Flask, Node.js, React, PostgreSQL, Oracle, MySQL**
**Programming & Tools: Java, Scala, Python, JavaScript, C++, Bash, Git, Jenkins, GitHub Actions, Bitbucket**

## Professional Experience

**FirstEigen LLC** Mar 2025 – Present

*AI Technical Architect | Naperville, IL(US)*

- Owned the backend architecture for data-matching pipeline, redesigning Spark pipelines with hash-based joins and Bloom-filter pruning to stabilize performance under high-volume workloads, gaining match throughput 3.7×.
- Diagnosed and fixed systemic backend issues: inconsistent partition strategies, skew-heavy joins, inefficient data materialization, and runaway cluster autoscaling, reducing compute spend by 28%.
- Took full operational ownership of Spark-based data-quality pipelines, implemented monitoring, lineage tracking, auto-recovery logic, and rollback-safe deployments, eliminating recurring ETL failures and reducing weekly support load by 40%.
- Led the architecture of Databuck's Agentic AI MCP Server capable of rule generation, LLM-assisted tool execution, and RAG-based help automation, increasing data-validation throughput 45% and reducing orchestration overhead by 35%.

**Deutsche Bank** Jul 2019 – Aug 2023

*Associate (Big Data & Cloud) | Pune, India*

- Designed and engineered a Spark-Scala decision engine on GCP as part of regulatory risk-computation backend, increasing risk computation efficiency by 79%, reducing runtime by 68% and saving $3M annually.
- Prototyped streaming inference for gradient-boosted risk models on Spark Structured Streaming, implementing sliding-window feature updates and adaptive thresholds that improved precision-recall balance 22%.
- Automated compliance-reporting via Cloud Run and BigQuery, solving historical bottlenecks like inconsistent data-hydration windows and bottlenecked cron execution, reducing 70% manual intervention and recovering $1M in yearly revenue.
- Troubleshot and resolved systemic production issues across terabyte-scale Spark jobs: schema mismatches, silent precision loss in financial metrics, intermittent GCS read stalls, and executor OOM failures caused by misaligned caching logic.
- Migrated 60+ legacy Hadoop pipelines to a modern Spark architecture, replacing brittle XML-based configs with code-driven transformations and dynamic partition layouts, resulting 40% lower runtime and significantly fewer reprocessing incidents.
- Built an in-house job scheduler using Cloud Functions, Airflow, and Spring Boot, removing $1 M in licensing costs and saving 40+ dev hours per month.
- Refactored legacy ETL pipelines with modular Spark transformations and dynamic partitioning, improving maintainability and reducing compute cost 25%.

## Education

**Purdue University** *Master of Science in Computer Science* ***(GPA: 4.00 / 4.00)*** Aug 2023 - May 2025
**Jadavpur University** *BE in Electronics and Tele-Communication* ***(GPA: 8.00 / 10.00)*** Jul 2015 – May 2019

## Research

***Research Assistant | Purdue University (West Lafayette, IN)*** Aug 2024 – May 2025

Developed "Cardi-GPT" (arXiv:2510.24737), an end-to-end diagnostic agent that processes 12-Lead ECG Signals via 16-block ResNet and provides clinical interpretations through a conversational interface.